QR CODE **WATCH THE VIDEO BEFORE YOU SOLVE** YouTube • Lesson 20

RULE · Area formulas (maydon)

Shape	Formula	Variables
Rectangle	$A = lw$	l=length, w=width
Triangle	$A = \frac{1}{2}bh$	b=base, h=height
Parallelogram	$A = bh$	b=base, h=height
Trapezoid	$A = \frac{1}{2}(a+b)h$	a, b=parallel sides
Circle	$A = \pi r^2$	r=radius

If given the **diameter**, halve it first: $r = d/2$. $d = 18 \Rightarrow r = 9 \Rightarrow A = 81\pi$, not 324π.

RULES SHEET – 2/2

RULE · Volume formulas (hajm)

Solid	Formula	Note
Rectangular prism	$V = lwh$	product of 3 dimensions
Cylinder	$V = \pi r^2 h$	base area × height
Cone	$V = \frac{1}{3}\pi r^2 h$	one-third of same-r, h cylinder
Sphere	$V = \frac{4}{3}\pi r^3$	depends only on r – no height

Never drop the $\frac{1}{3}$ (cone) or $\frac{4}{3}$ (sphere) – a cone is exactly one-third of the matching cylinder's volume.

$r = 6$: $V = \frac{4}{3}\pi(6)^3 = \frac{4}{3}\pi(216) = 288\pi$

RULE · Composite figures: add or subtract

Pieces joined together → **add**. A piece cut out/a hole → **subtract**.

Joined: 10×6 rect. $+ 4 \times 3$ rect. $= 60 + 12 = 72$

Hole: 10×10 square, radius-3 hole $= 100 - 9\pi$ (not $100 + 9\pi$)

Sanity check: subtracting a piece must give a result smaller than the whole, never larger.